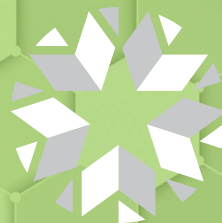


OKLAHOMA SCHOOL TESTING PROGRAM

TEST BLUEPRINT AND ITEM SPECIFICATIONS
GRADE 8 SCIENCE



OKLAHOMA
Education

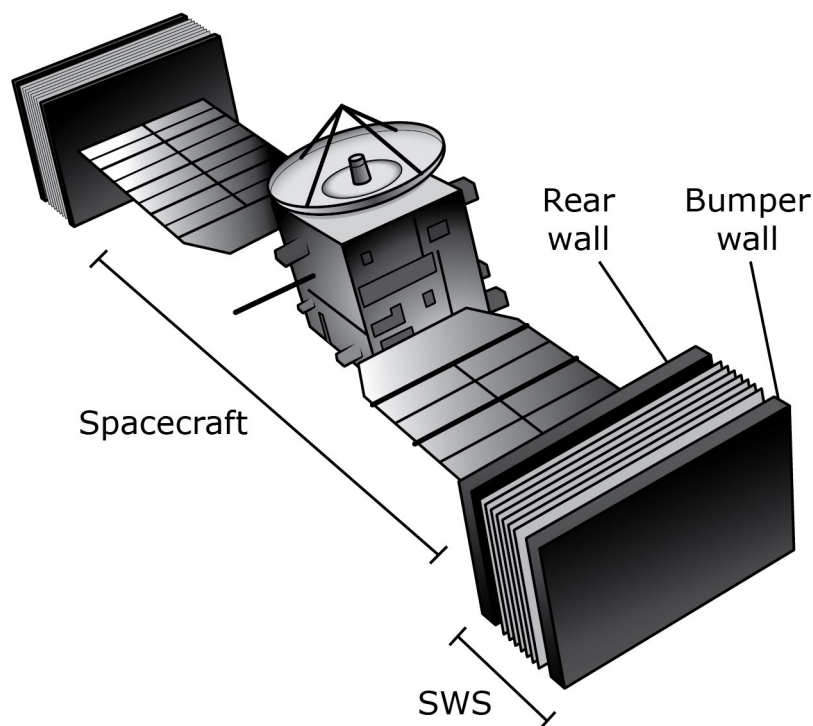
Sample Cluster for 8.PS2.1:

Use the information to answer the following questions.

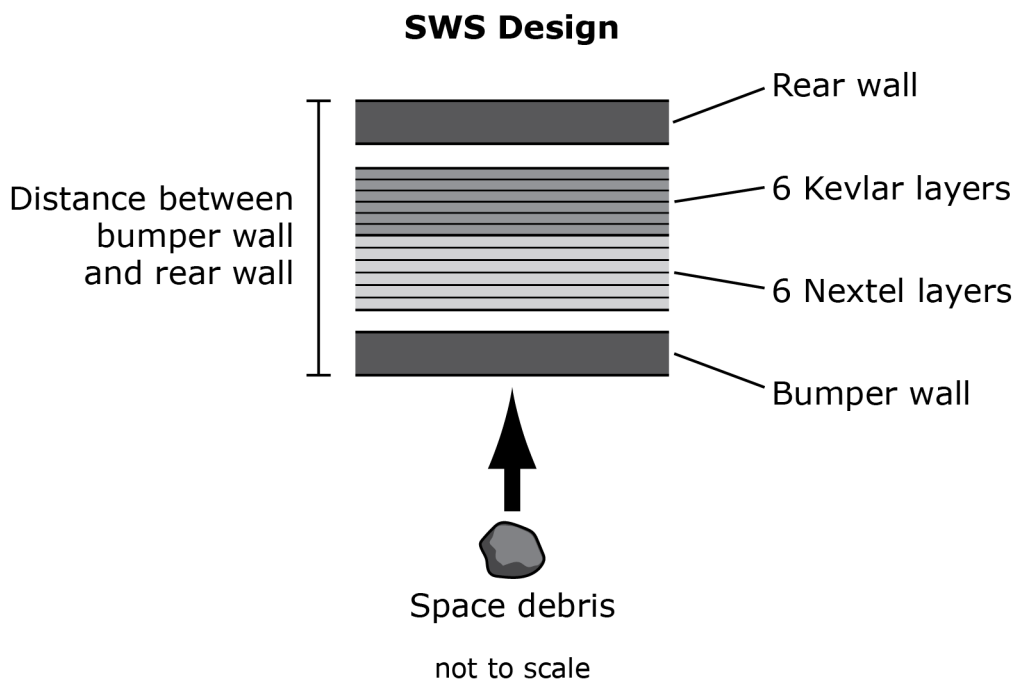
A student is reading about spacecraft in class. They are surprised to learn that as a spacecraft orbits Earth there are small objects in space called debris that can collide with the larger spacecraft and cause damage. They research some information about how scientists have tried to address this problem.

The student learns from their research that to help decrease damage to a spacecraft, scientists designed shields to protect the spacecraft from collisions with space debris. One type of shield is called the Stuffed Whipple Shield (SWS). The diagram shows the SWS attached to a spacecraft.

SWS Attached to Spacecraft



The SWS consists of an aluminum bumper wall, layers of material called Nextel and Kevlar, and an aluminum rear wall closest to the spacecraft. The student also finds a diagram that shows a close-up of the SWS design and how space debris collides with the bumper wall of the SWS.



Scientists ran a simulation building different SWS designs. The data table lists the bumper wall thickness, rear wall thickness, and distance between the bumper wall and rear wall in centimeters (cm) for four SWS versions.

SWS Versions Data Table

SWS Version	Bumper Wall Thickness (cm)	Rear Wall Thickness (cm)	Distance between Bumper Wall and Rear Wall (cm)
W	0.20	0.20	4
X	0.20	0.20	3
Y	0.25	0.25	4
Z	0.25	0.25	3

(Items on the following pages)

1

Based on the SWS Versions Data Table, the student claims that SWS Version Y would be most effective at reducing damage to the spacecraft's rear wall when space debris collides with the SWS bumper wall.

Which statement supports the student's claim?

- A** The collision with thicker walls would break the debris into smaller pieces and increase the speed of the debris as it travels a greater distance from the bumper wall to the rear wall.
- B** The collision with thicker walls would break the debris into smaller pieces and reduce the speed of the debris as it travels a greater distance from the bumper wall to the rear wall.
- C** The collision with thicker walls would break the rear wall into pieces, while the debris stays intact, and reduce the speed of the debris as it travels a greater distance from the bumper wall to the rear wall.
- D** The collision with thicker walls would break the rear wall into pieces, while the debris stays intact, and increase the speed of the debris as it travels a greater distance from the bumper wall to the rear wall.

Standard: Apply Newton's Third Law to design a solution to a problem involving the motion of two colliding objects in a system.

Cognitive Complexity:

SEP: Medium

DCI: High

CCC: Low

Overall Cognitive Complexity: Low Support

Rationale:

- A. Smaller pieces of debris would have a smaller mass, causing a decrease in speed.
- B. KEY: Breaking up the debris into smaller pieces (smaller mass) and reducing the speed of the debris would reduce the net force of the collision.**
- C. The collision with thicker walls would break any remaining debris into pieces instead of breaking the rear wall, and any intact debris would increase in speed.
- D. The speed of the debris would decrease, not increase, because the collision with thicker walls would break any remaining debris into pieces instead of breaking the rear wall.

- 2** Based on the information, which statement describes how an SWS should be designed so that it **best** shields the spacecraft from a collision with space debris?
- A** Decrease the thickness of the rear wall so that the debris has less material to contact when it collides with the SWS.
 - B** Increase the amount of material between the bumper wall and rear wall to reduce the force or forces in the collision.
 - C** Decrease the number of Kevlar layers so that the bumper wall and the rear wall are closer together when the debris collides with the SWS.
 - D** Increase the mass of the bumper wall and rear wall to be similar to that of typical debris so that the force or forces in the collision are balanced.

Standard: Apply Newton's Third Law to design a solution to a problem involving the motion of two colliding objects in a system.

Cognitive Complexity:

SEP: Medium

DCI: Medium

CCC: Medium

Overall Cognitive Complexity: Moderate Support

Rationale:

- A. Decreasing the thickness of the rear wall would limit how much the shield reduces the acceleration of any remaining solid debris, so this would not help shield the spacecraft from a collision with debris.
- B. KEY: Adding more Nextel and Kevlar layers between the bumper wall and rear wall would improve how much the shield reduces the acceleration of the debris, so that the force of the collision would be reduced and therefore rear wall damage would be reduced.**
- C. Reducing the amount of material that the space debris would collide with would limit how much the shield reduces the acceleration of the debris, so this would not be an improvement to reduce rear wall damage.
- D. Students may have the misconception that action-reaction forces can balance each other out, when this is not the case since these forces are between two different objects (the debris and the walls of the SWS).

- 3** Which statement **best** describes what scientists should consider when designing SWS devices?
- A** If multiple versions of SWS devices are built, there can be many trials in which an object collides with a device.
 - B** After SWS devices collide with an object, the devices must be destroyed before any more devices can be built.
 - C** The first of several SWS devices built is the most effective at protecting objects from being damaged during a collision between an object and the device.
 - D** Building SWS devices with the most expensive materials will make those devices most effective at reducing damage from a collision with an object.

Standard: Apply Newton's Third Law to design a solution to a problem involving the motion of two colliding objects in a system.

Cognitive Complexity:

SEP: Low

DCI: Low

CCC: Low

Overall Cognitive Complexity: Moderate Support

Rationale:

- A. KEY: SWS device design is an iterative process, where scientists consider alternate design pathways to improve the effectiveness of the SWS device.**
- B. Scientists develop SWS devices through the design process so they can revisit how earlier devices were built and refine the design of the device instead of starting from scratch with a new design.
- C. Scientists designing an SWS device work to improve upon their initial designs, rather than assuming that the first solution is the best solution.
- D. The cost of an SWS device is a design constraint, as space missions typically operate within strict budget limits.

Use the information to answer the following questions.

A Green Team and a Blue Team prepare for a shopping cart race. They carry out an investigation that will help them so they can do well in the race. They used a spring scale to measure the force in newtons (N = newtons). The procedure and a diagram of the experimental setup are shown.

Procedure:

1. Place 25 kg of groceries in a shopping cart.
2. Attach a rope and a spring scale to the front of the shopping cart.
3. Have a runner pull the cart 10 meters across a parking lot as fast as they can.
4. Use the spring scale to measure the amount of force, in newtons, the runner applies to the cart while pulling.
5. Measure the time it takes for the runner to pull the cart 10 meters.
6. Use the time and distance to calculate the speed of the runner.
7. Repeat the steps for each runner.



(continued on next page)

The data collected during the investigation are shown in the table.

Team	Runner	Pull Force (N)	Time it took to travel 10 meters (s)
Green	1	8	5.6
Green	2	3.3	8.7
Green	3	14.2	4.2
Green	4	5.1	7.0
Blue	5	4.7	7.3
Blue	6	4.2	7.7
Blue	7	3.9	8.0
Blue	8	5.4	6.8

- 4** The teams decided to investigate the effects of mass on the force needed to pull the shopping cart.

What could the students change in the investigation to make the cart go faster?

- A** Put more mass in the cart.
- B** Put less mass in the cart.
- C** Use a longer rope to pull the shopping cart.
- D** Use a larger shopping cart.

Standard: Plan an investigation to provide evidence that the change in an object's motion depends on the sum of the forces on the object and the mass of the object.

Cognitive Complexity:

SEP: Medium

DCI: Medium

CCC: Medium

Overall Cognitive Complexity: Moderate Support

Rationale:

- A. Putting more mass in the cart will make the cart heavier and thereby harder to pull making it slower.
- B. KEY: Putting less mass in the cart will make the cart lighter and therefore go faster.**
- C. Using a different rope will not make the cart lighter and thereby easier to pull.
- D. The shopping carts will have the same relative mass thereby this would not change the outcome of the investigation.

- 5** The students noticed the surface they were racing on had different sized rocks on it. They wondered if the rocks being in the way would affect their race.

Which explanation **best** describes how the rocks will affect the shopping cart?

- A** Only the big rocks will affect the shopping cart because the force of the rocks will be going in the same direction as the shopping cart's wheels when it hits them.
- B** Only the small rocks will affect the shopping cart because the force of the rocks will be going in the same direction as the shopping cart's wheels when it hits them.
- C** None of the rocks will affect the shopping cart because the rocks have zero force.
- D** All the rocks will affect the shopping cart because the force of the rocks will be going in the opposite direction as the shopping cart's wheels when it hits them.

Standard: Plan an investigation to provide evidence that the change in an object's motion depends on the sum of the forces on the object and the mass of the object.

Cognitive Complexity:

SEP: Medium

DCI: High

CCC: Medium

Overall Cognitive Complexity: Low Support

Rationale:

- A. The big rocks will affect the shopping cart but so will the small rocks because the rocks have an opposite net force as the shopping cart's wheels upon impact.
- B. The small rocks will affect the shopping cart but so will the big rocks because the rocks have an opposite net force to the shopping cart's wheels upon impact.
- C. The shopping cart could roll over some of the rocks but all the rocks will slow down the shopping cart because the rocks have an opposite net force to the shopping cart's wheels upon impact.
- D. KEY:** All the rocks will affect the shopping cart because the rocks have an opposite net force to the shopping cart's wheels upon impact.

- 6** Based on the data in the chart which two runners should the Green Team select to win the race and why?
- A** Runners 2 and 3, because together they have the lowest pull force and slowest time.
 - B** Runners 1 and 3, because together they have the highest pull force and fastest time.
 - C** Runners 4 and 1, because together they have the lowest pull force and fastest time.
 - D** Runners 2 and 4, because together they have the highest pull force and slowest time.

Standard: Plan an investigation to provide evidence that the change in an object's motion depends on the sum of the forces on the object and the mass of the object.

Cognitive Complexity:

SEP: Medium

DCI: High

CCC: Low

Overall Cognitive Complexity: Low Support

Rationale:

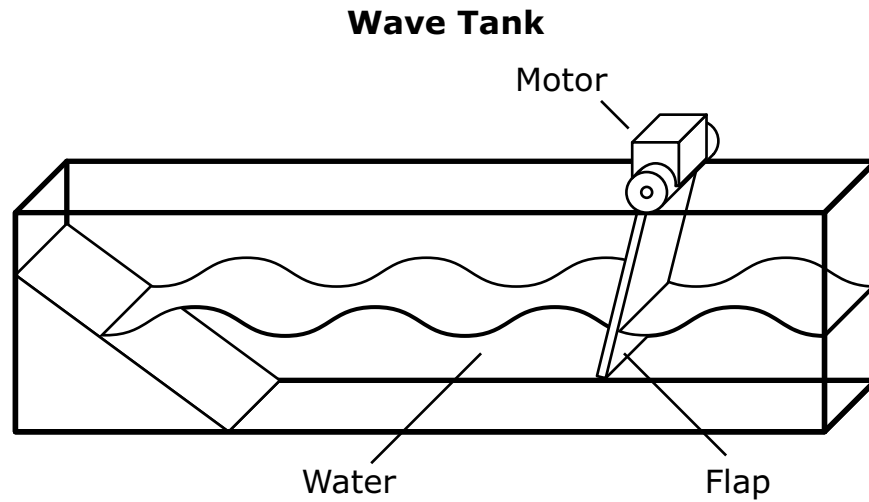
- A. Runners 2 and 3 have a combined pull force of 17.5 N and an average time of 6.45 s.
- B. KEY: Runners 1 and 3 have a combined pull force of 22.2 N and an average time of 4.9 s, making them the fastest and the strongest pull force.**
- C. Runners 4 and 1 have a combined pull force of 13.1 N and an average time of 6.3 s.
- D. Runners 2 and 4 have a combined pull force of 8.4 N and an average time of 7.85 s.

Sample Cluster for 8.PS4.1:

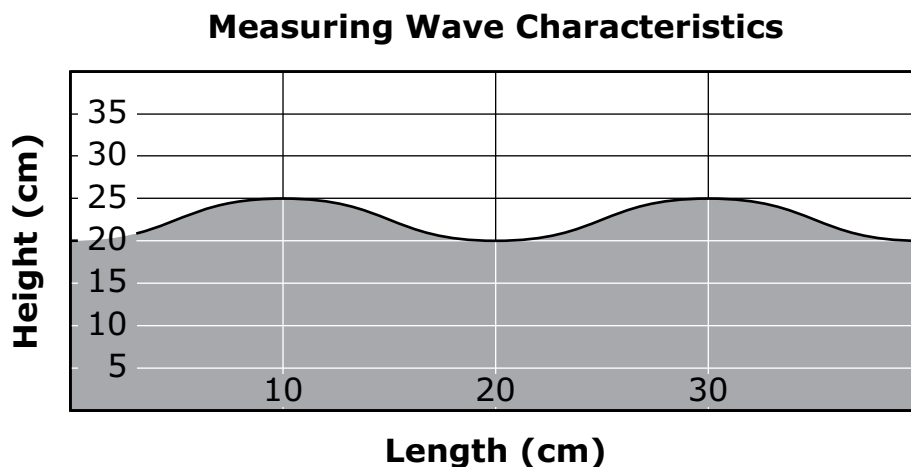
Use the information to answer the following questions.

A student observed small waves on a lake. The next day, the waves on the lake were two times higher. The student wonders why the wave height changed.

The student plans an investigation using a wave tank. A wave tank is a large, clear plastic container filled with water. A flap at one end of the tank can be moved back and forth by a motor. This motion creates waves in the tank. The diagram shows a wave tank.



The student can measure characteristics of the waves created in the tank. The diagram shows a wave in the tank. The tank has scales marked on the side and bottom.



(Items on the following pages)

- 7** The student studies the Measuring Wave Characteristics diagram to find the amplitude of the wave in the tank. The student wants to draw a line showing where the amplitude of the wave is.

Based on the diagram, where should the amplitude of the wave be drawn?

- A** The amplitude line should be drawn halfway between the 0 and 25 cm lines on the y -axis.
- B** The amplitude line should be drawn at the 20 cm line on the y -axis.
- C** The amplitude line should be drawn halfway between the 20 and 25 cm lines on the y -axis.
- D** The amplitude line should be drawn at the 25 cm line on the y -axis.

Standard: Use mathematical representations to describe a simple model for waves that includes how the amplitude of a wave is related to the energy in a wave.

Cognitive Complexity:

SEP: Med

DCI: Med

CCC: Low

Overall Cognitive Complexity: Moderate Support

Rationale:

- A. This represents the height of the crest but the amplitude is found by looking at the difference in the trough height and the crest height and dividing it in half.
- B. This height represents the trough, but the amplitude is found by looking at the differences in the trough height and the crest height and dividing it in half.
- C. KEY: From trough to crest the wave is 5.0 cm. The amplitude would be half of the total height, which is 2.5 cm.**
- D. This represents the height of the crest but the amplitude is found by looking at the difference in the trough height and the crest height and dividing it in half.

- 8** The student thinks about the waves on the lake being two times higher than the day before.

How could the student cause the waves in the wave tank to become two times higher?

- A** Change the energy input from the motor to one-quarter the original amount.
- B** Change the energy input from the motor to one-half the original amount.
- C** Change the energy input from the motor to two times the original amount.
- D** Change the energy input from the motor to four times the original amount.

Standard: Use mathematical representations to describe a simple model for waves that includes how the amplitude of a wave is related to the energy in a wave.

Cognitive Complexity:

SEP: Med

DCI: High

CCC: Low

Overall Cognitive Complexity: Low Support

Rationale:

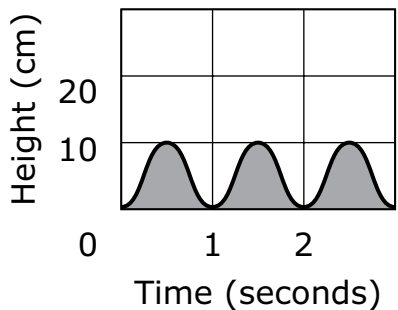
- A. This is taking the energy of the wave and making it reversely proportional to the square of the amplitude.
- B. This is taking the energy of the wave and making it reversely proportional to the amplitude.
- C. This is taking the energy of the wave and making it proportional to the amplitude.
- D. KEY: The energy of the wave is proportional to the square of the amplitude. If the height of a water wave is doubled, each wave will have four times the energy.**

This question has two parts. Be sure to answer both parts of the question.

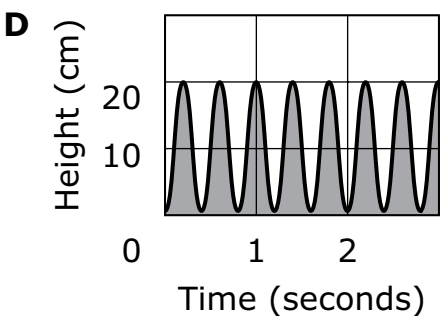
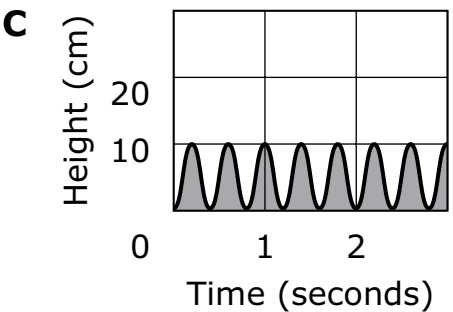
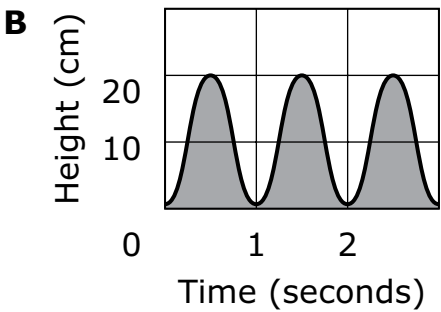
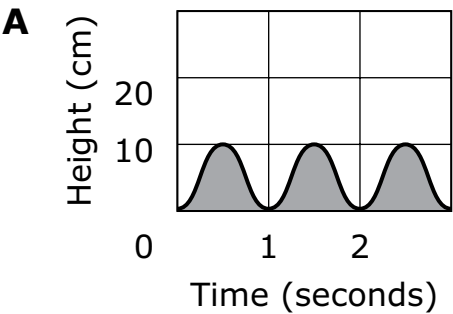
9 Part A

The student wants to make two wave models. One wave model will represent a decrease in wavelength without changing the amplitude. The other wave model will represent an increase in frequency with a change in the amplitude.

The diagram represents the original wave.

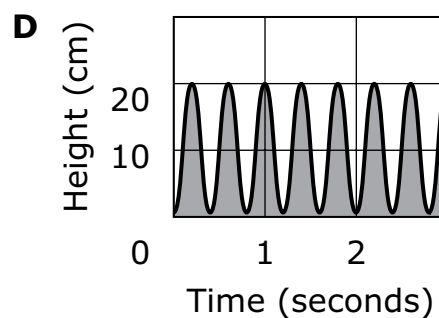
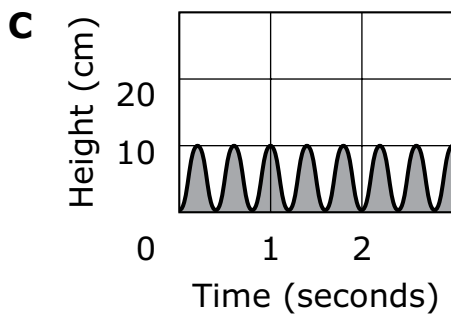
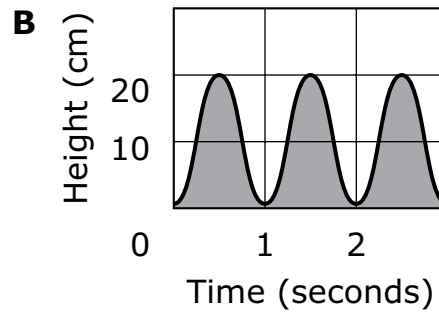
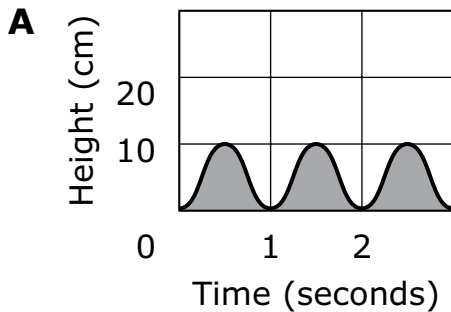


Which diagram represents a decrease in wavelength for the original wave without a change in amplitude?



Part B

Which diagram represents an increase in frequency for the original wave with a change in amplitude?



Standard: Use mathematical representations to describe a simple model for waves that includes how the amplitude of a wave is related to the energy in a wave.

Cognitive Complexity:

SEP: High

DCI: High

CCC: Low

Overall Cognitive Complexity: Low Support

Rationale:

Part A

- A. This wave model does not change the wavelength or the amplitude.
- B. The wavelength stays the same but the amplitude is changed in this wave.
- C. **KEY:** The amplitude is the same because it is the same height and the wavelength decreases because the distance between crests is closer together than the original wave.
- D. Both the wavelength and the amplitude has changed for this wave.

Part B

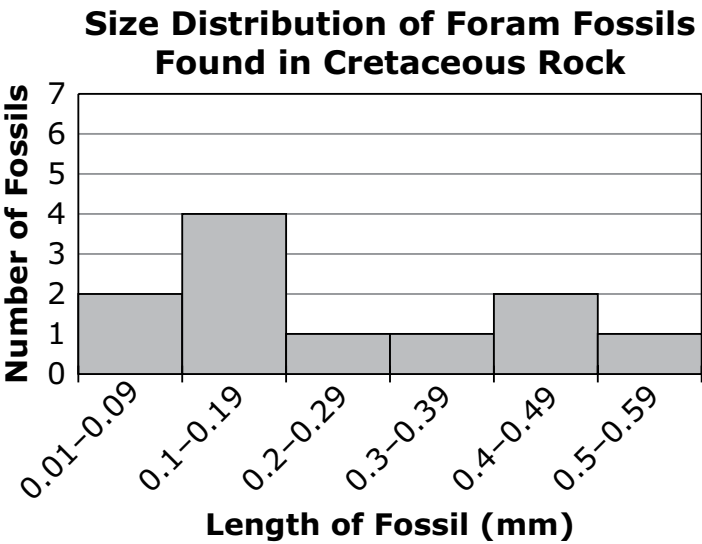
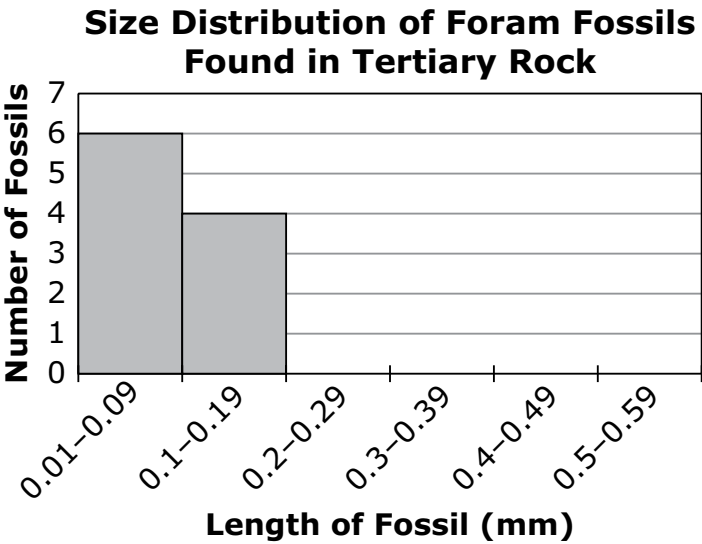
- A. This wave model does not change the frequency or the amplitude.
- B. The frequency stays the same but the amplitude is changed in this wave.
- C. The amplitude is the same because it is the same height and the frequency increases because the number of crests in 2 seconds is more than the original wave.
- D. **KEY:** Both the frequency and the amplitude has changed for this wave.

Sample Cluster for 8.LS4.1:

Use the information to answer the following questions.

To answer questions about Earth’s history, students looked at rock samples from different layers of the ocean floor. The rock samples contained fossils of very small ocean organisms called forams.

Some of the rock formed during the Cretaceous time period from 145.5 to 65.5 million years ago, before an event called the K-T extinction. The rest of the rock formed during the Tertiary time period from 65.5 to 35.4 million years ago, after the K-T extinction. In their investigation, the students measured the lengths of the foram fossils in the rock samples. The graphs show the size ranges of the foram fossils in rock samples from each time period.



- 10** Which statement is supported by the data in the two graphs?
- A** Forams of all sizes had more food sources in the Cretaceous period.
 - B** Forams with a larger size had more predators in the Cretaceous period.
 - C** Forams with a smaller size had a survival advantage in the Tertiary period.
 - D** Forams that were mid-sized had a higher reproductive rate in the Tertiary period.

Standard: Analyze and interpret data for patterns in the fossil record that document the existence, diversity, extinction, and change of life forms throughout the history of life on Earth.

Cognitive Complexity:

SEP: High

DCI: High

CCC: Low

Overall Cognitive Complexity: Low Support

Rationale:

- A. The student may think that the greater variety of Cretaceous forams indicates that were more food sources.
- B. The student may think that the greater variety of Cretaceous forams indicates that were more predators.
- C. KEY: The average foram size decreased in the Tertiary period, which indicates that smaller size was an advantage.**
- D. The student may read the Cretaceous graph and may think the data reflects reproductive rates.

- 11** Based on the information, how are the foram fossils from the two time periods different?
- A** The Cretaceous fossils are found in more locations than the Tertiary fossils.
 - B** The Cretaceous fossils show a greater variety of body size than the Tertiary fossils.
 - C** The Cretaceous fossils were more successful than Tertiary fossils at surviving the K-T extinction.
 - D** The Cretaceous fossils became more common and the Tertiary fossils disappeared from the fossil record.

Standard: Analyze and interpret data for patterns in the fossil record that document the existence, diversity, extinction, and change of life forms throughout the history of life on Earth.

Cognitive Complexity:

SEP: Med

DCI: Low

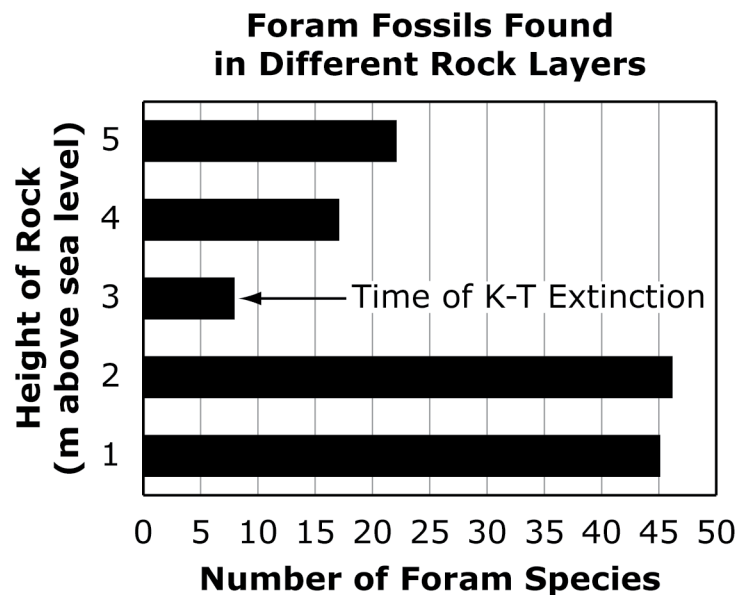
CCC: Low

Overall Cognitive Complexity: Moderate Support

Rationale:

- A. The student may think that the different rocks come from different locations.
- B. KEY: The Tertiary graph shows sizes only from 0.01–0.19 mm, while the Cretaceous graphs shows sizes from 0.01–0.59 mm.**
- C. The student may be confused regarding the timing of the K-T extinction.
- D. The student may think that the zero values in the graph indicate that the fossils disappeared.

- 12** The students also gathered data about the number of fossilized foram species in rock layers at different heights above sea level from the Cretaceous and Tertiary time periods.



What is the **most likely** reason for the difference in the number of foram species in the layers from 3 to 5 meters above sea level?

- A** Individual forams became different species until the time of the K-T extinction.
- B** The number of foram species decreased steadily until the time of the K-T extinction.
- C** The surviving foram species diversified to fill the available habitats after the K-T extinction.
- D** Several foram species moved from other habitats in the world to this location after the K-T extinction.

Standard: Analyze and interpret data for patterns in the fossil record that document the existence, diversity, extinction, and change of life forms throughout the history of life on Earth.

Cognitive Complexity:

SEP: High

DCI: High

CCC: Med

Overall Cognitive Complexity: Low Support

Rationale:

- A. The student may think that individuals become different species.
- B. The student may think that moving down from the surface is going forward in time rather than in reverse.
- C. **KEY:** The number of species increases from layers 3–5 indicating that diversification occurred.
- D. The student may think that migration is the most likely explanation.

Use the information to answer the following questions.

A student is cooking with a type of plant called thyme and notices that thyme has a fragrant smell. The student wonders how this smell is produced and does some research. The student learns that the smell produced by thyme is due to oils that the plant produces. Some types of thyme plants produce oils with different chemicals that affect the plant's smell.

Thyme plants that produce oils with stronger-smelling chemicals, like phenols, deter predators from eating the thyme plants. However, having this phenol trait also makes these plants more likely to perish from freezing temperatures than thyme plants without phenols.

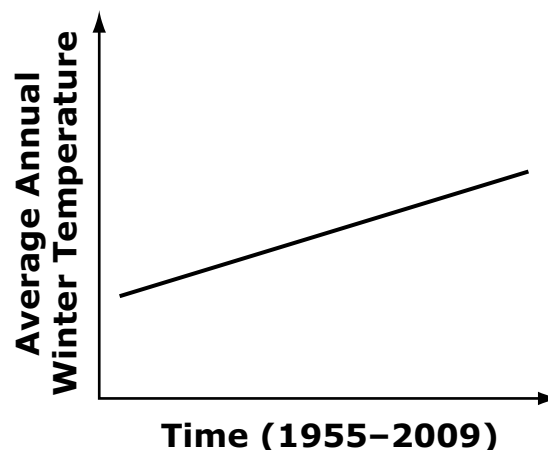
The student finds a data table that shows the percentage of thyme plants with the phenolic trait and nonphenolic trait in 1974 and 2010 in a region of France where these thyme plants grow.

Phenolic and Nonphenolic Trait Percentage in Thyme Population, 1974 and 2010

Year	Percentage of Thyme Population	
	Phenolic Trait	Nonphenolic Trait
1974	0	100
2010	44	56

The student also finds a graph that shows how the average annual winter temperature changed over time from 1955–2009 in this region of France where thyme plants grow.

Average Annual Winter Temperature in Environment with Thyme Plants, 1955–2009



13 Which conclusion could the student make that is supported by the data?

- A** From 1974–2010, thyme plants with the phenolic trait were more likely to be eaten than before 1974.
- B** From 1974–2010, thyme plants changed their nonphenolic trait to a phenolic trait over their lifetimes.
- C** From 1974–2010, thyme plants with the phenolic trait were less likely to freeze to death than before 1974.
- D** From 1974–2010, thyme plants with the phenolic trait were stronger than thyme plants with the nonphenolic trait.

Standard: Use mathematical representations to support explanations of how natural selection may lead to increases and decreases of specific traits in populations over time.

Cognitive Complexity:

SEP: Low

DCI: Medium

CCC: Medium

Overall Cognitive Complexity: Low Support

Rationale:

- A. This is not supported by the data because average annual winter temperature increased by 2010 and corresponded to an increase in the distribution of the phenolic trait in the thyme population, which suggests that individuals with this trait were surviving to reproduce.
- B. Students may have the misconception that individuals can change their traits over their lifetimes, rather than the distribution of traits changing in a population over time due to natural selection.
- C. KEY: The percentage of thyme plants with the phenolic trait increased relative to the nonphenolic trait during the same time period when average annual winter temperatures increased, suggesting that the freezing risk was lowered enough for the phenolic trait to be beneficial for reducing predation risk in the thyme plant population.**
- D. Students may have the misconception that natural selection is driven by individuals being strong enough to survive (“survival of the fittest”) and that this explains why both phenolic and nonphenolic thyme plants are present in the 2010 thyme population.

- 14** Based on the data, which statement explains the distribution of the nonphenolic trait in the thyme population from 1974–2010?
- A** Thyme plants with the nonphenolic trait rapidly changed to gain the phenolic trait because the environment became colder over time.
 - B** Thyme plants with the nonphenolic trait did not use their ability to withstand freezing temperatures because the environment became warmer over time.
 - C** The nonphenolic trait was more likely to be inherited by thyme plants over many generations because the nonphenolic trait is advantageous in a colder environment.
 - D** The nonphenolic trait was less likely to be inherited by thyme plants over many generations because the nonphenolic trait is not advantageous in a warmer environment.

Standard: Use mathematical representations to support explanations of how natural selection may lead to increases and decreases of specific traits in populations over time.

Cognitive Complexity:

SEP: Low

DCI: Medium

CCC: Medium

Overall Cognitive Complexity: Low Support

Rationale:

- A. Students may have the misconception that organisms can change their features to suit their environment.
- B. Students may have the misconception that traits that are not used become less frequent in a population over time.
- C. Students may not recognize that the phenolic trait is disadvantageous in a colder environment.
- D. KEY:** Since the nonphenolic trait helps thyme plants survive freezing temperatures, the increase in average winter temperatures from 1974–2010 likely explains why the frequency of nonphenolic thyme plants decreased, since nonphenolic thyme plants are at greater risk of being eaten by predators than phenolic thyme plants.

- 15** Based on the data, which table represents how the proportion of thyme plants with each trait would **most likely** change if the average annual winter temperature steadily increased from 2010 to 2050?

A **Percentage of Thyme Population in 2050**

Phenolic Trait	Nonphenolic Trait
Less than 44	More than 56

B **Percentage of Thyme Population in 2050**

Phenolic Trait	Nonphenolic Trait
More than 44	More than 56

C **Percentage of Thyme Population in 2050**

Phenolic Trait	Nonphenolic Trait
Less than 44	Less than 56

D **Percentage of Thyme Population in 2050**

Phenolic Trait	Nonphenolic Trait
More than 44	Less than 56

(continued on next page)

Standard: Use mathematical representations to support explanations of how natural selection may lead to increases and decreases of specific traits in populations over time.

Cognitive Complexity:

SEP: High

DCI: Medium

CCC: Medium

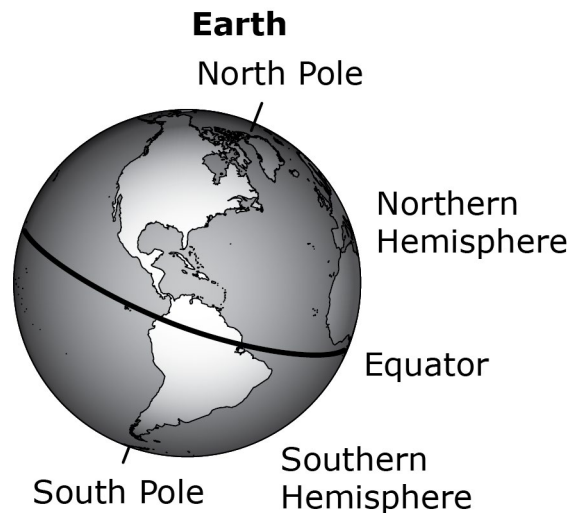
Overall Cognitive Complexity: Low Support

Rationale:

- A. If average annual winter temperature continued to increase, it's likely that phenolic thyme plants that can deter predators would be more likely to survive and reproduce than nonphenolic thyme plants, so the frequency of phenolic thyme plants in the population would likely increase while nonphenolic thyme plants would decrease.
- B. If average annual winter temperatures continued to increase, it's likely that phenolic thyme plants that can deter predators would be more likely to survive and reproduce than nonphenolic thyme plants, so the frequency of nonphenolic thyme plants in the population would likely decrease, not increase.
- C. If average annual winter temperatures continued to increase, it's likely that phenolic thyme plants that can deter predators would be more likely to survive and reproduce than nonphenolic thyme plants, so the frequency of phenolic thyme plants in the population would likely increase, not decrease.
- D. **KEY:** Based on the frequency of thyme plants with the phenolic trait increasing and the nonphenolic trait decreasing with an increase in average annual winter temperature, a further increase in average winter temperatures would be likely to further provide phenolic thyme plants with a survival advantage against predation.

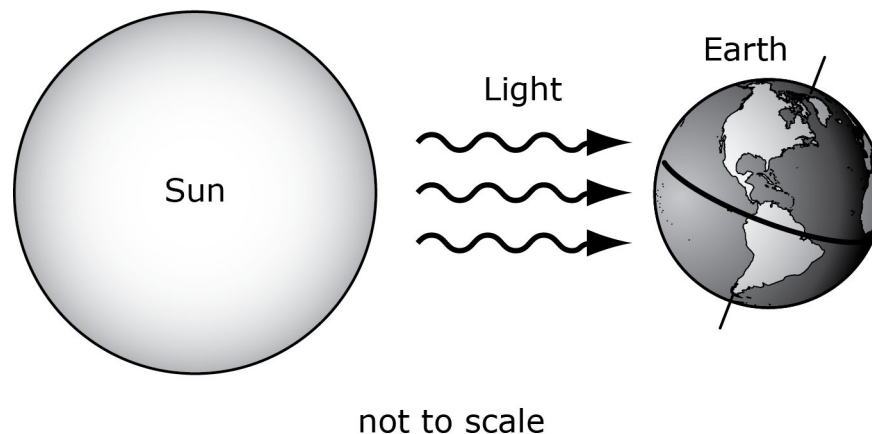
Use the information to answer the following questions.

A student is indoors at school on a winter day in Oklahoma when they are surprised to learn from their teacher that in some parts of South America it is summertime. They ask their teacher why the seasons are different between these locations and their teacher shows them a model of Earth in relation to the Sun. Their teacher explains that Earth has a North Pole and a South Pole. Earth also has two different regions, or hemispheres: a Northern Hemisphere above the equator and a Southern Hemisphere below the equator. The diagram shows the equator and the poles and hemispheres of Earth.



Oklahoma is in the Northern Hemisphere of Earth, while some parts of South America are in the Southern Hemisphere. The model shows the position of Earth relative to the Sun when it is December.

**Position of Earth in Orbit Around Sun
in December**



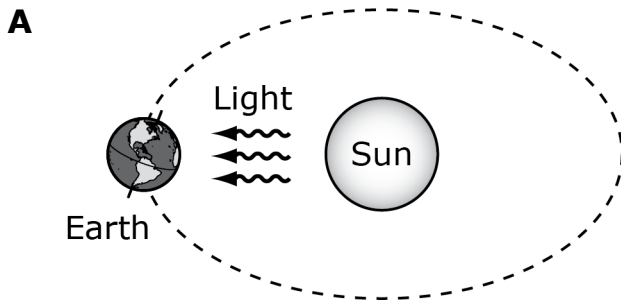
(continued on next page)

The student also finds information about seasons at different times of the year in relation to Earth’s hemispheres.

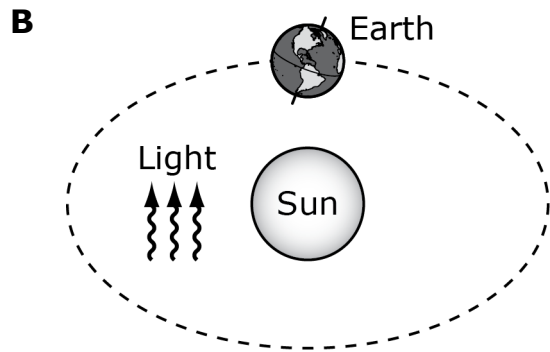
Season by Month and Hemisphere

Month	Season	Hemisphere
December	Winter	Northern
December	Summer	Southern
March	Spring	Northern
March	Fall	Southern
June	Summer	Northern
June	Winter	Southern
September	Fall	Northern
September	Spring	Southern

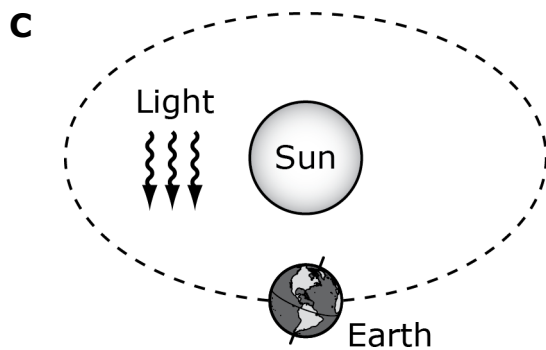
- 16** Based on the Position of Earth in Orbit Around Sun in December model, which model **best** represents the position of Earth relative to the Sun in June?



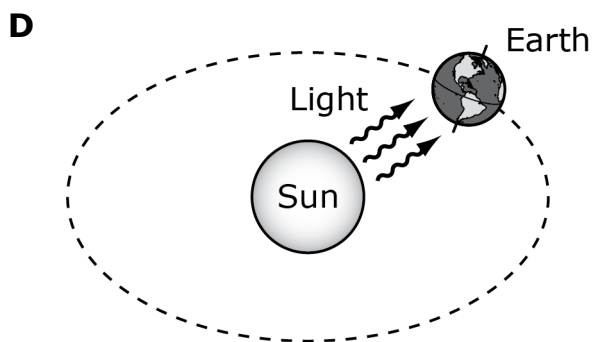
The model shows the Sun shining mostly on the Northern Hemisphere of Earth, with Earth's North Pole tilted toward the Sun.



The model shows the Sun shining on both of Earth's hemispheres with Earth directly behind the Sun.



The model shows the Sun shining on both of Earth's hemispheres with Earth directly in front of the Sun.



The model shows the Sun shining mostly on the Southern Hemisphere of Earth, with Earth's South Pole tilted toward the Sun.

(continued on next page)

Standard: Develop and use a model of the Earth-Sun-Moon system to describe the cyclic patterns of lunar phases, eclipses of the Sun and Moon, and seasons.

Cognitive Complexity:

SEP: Medium

DCI: Medium

CCC: High

Overall Cognitive Complexity: Moderate Support

Rationale:

- A. **KEY:** This model correctly represents the position of Earth and the Sun in June because at this time Earth's North Pole is tilted toward the Sun and Earth's South Pole is tilted away from the Sun.
- B. This model does not represent the position of Earth and the Sun in June because it does not provide evidence that each hemisphere receives different intensities of sunlight.
- C. This model does not represent the position of Earth and the Sun in June because the model shows the Sun shining on both hemispheres rather than showing a difference in the intensity of sunlight each hemisphere receives.
- D. This model represents the position of Earth and the Sun at a time that better reflects the winter months on Earth since Earth's South Pole is tilted toward the Sun and Earth's North Pole is tilted away from the Sun.

- 17** Based on the Position of Earth in Orbit Around Sun in December model, which statement explains why winter occurs at different times of the year between hemispheres?
- A** Earth is relatively farther away from the Sun when winter occurs on Earth.
 - B** Earth orbits the Sun slowly so that throughout the year global temperatures are colder when winter occurs.
 - C** Earth has a tilted axis that affects when winter occurs on Earth due to Earth's relative position while orbiting the Sun throughout the year.
 - D** Earth spins relative to the Sun, which affects when winter occurs due to Earth facing the Sun in the same direction as it spins throughout the year.

Standard: Develop and use a model of the Earth-Sun-Moon system to describe the cyclic patterns of lunar phases, eclipses of the Sun and Moon, and seasons.

Cognitive Complexity:

SEP: Medium

DCI: Medium

CCC: Low

Overall Cognitive Complexity: Moderate Support

Rationale:

- A. Students may have the misconception that the seasons are caused by Earth's general proximity to the Sun (when Earth is far away from the Sun it is winter because it is colder).
- B. Students may have the misconception that the rate at which Earth orbits the Sun primarily affects the seasons throughout the year.
- C. KEY: Since Earth has a tilted axis as it orbits the Sun, there is a differential intensity of sunlight on different parts of Earth throughout the year.**
- D. Students may have the misconception that Earth's spin primarily affects the seasons throughout the year.

- 18** Which statement describes how the student could improve the Position of Earth in Orbit Around Sun in December model to explain the seasons on Earth in December?
- A** Show that the Southern and Northern Hemispheres receive high-intensity sunlight from the Sun.
 - B** Show that the Southern and Northern Hemispheres receive low-intensity sunlight from the Sun.
 - C** Show that the Sun shines lower-intensity light on the Southern Hemisphere than on the Northern Hemisphere.
 - D** Show that the Sun shines higher-intensity light on the Southern Hemisphere than on the Northern Hemisphere.

Standard: Develop and use a model of the Earth-Sun-Moon system to describe the cyclic patterns of lunar phases, eclipses of the Sun and Moon, and seasons.

Cognitive Complexity:

SEP: Medium

DCI: Medium

CCC: High

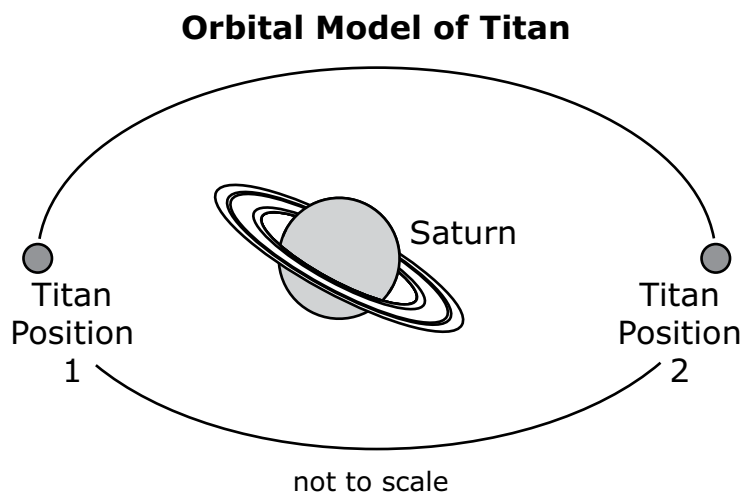
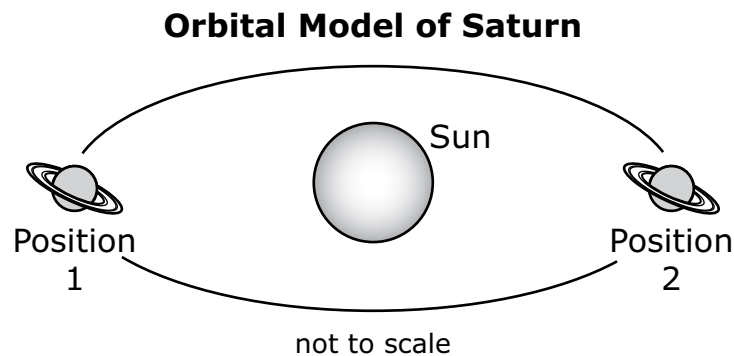
Overall Cognitive Complexity: Moderate Support

Rationale:

- A. This would indicate that both hemispheres are experiencing summer in December, but based on the table, the Northern Hemisphere is experiencing winter and the Southern Hemisphere is experiencing summer.
- B. This would indicate that both hemispheres are experiencing winter in December, but based on the table, the Northern Hemisphere is experiencing winter and the Southern Hemisphere is experiencing summer.
- C. Based on the table, this would better represent Earth in the month of June since it would reflect summer in the Northern Hemisphere and winter in the Southern Hemisphere.
- D. KEY: The table indicates that in December it is winter in the Northern Hemisphere and summer in the Southern Hemisphere, so the model should show that this is due to each hemisphere receiving different intensities of sunlight at this time of the year.**

Use the information to answer the following questions.

A class visits a planetarium where students watch a presentation that models movements in the solar system. During the presentation, the students see the planet Saturn and one of its moons, Titan. The students want to learn more about Saturn and Titan. They find models of Saturn’s orbit and Titan’s orbit. The models are shown.



The students also find a table that shows the mass of each object, as shown.

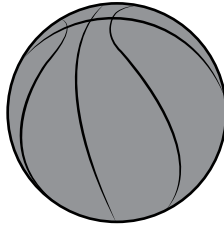
Mass of Three Objects in the Solar System

Object	Mass	Average Distance from the Sun	Average Distance from Saturn
Saturn	5.7×10^{26} kg	1.4×10^9 km	N/A
Sun	2.0×10^{30} kg	N/A	1.4×10^9 km
Titan	1.3×10^{23} kg	1.4×10^9 km	1.2×10^6 km

- 19** The students want to make a model to compare the size of the three objects in the table.

According to the data, which model **best** shows the difference in sizes of the objects?

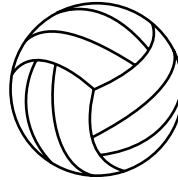
A



Sun

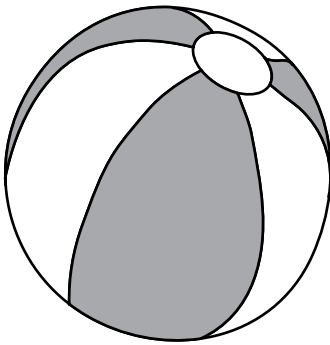


Saturn



Titan

B



Sun

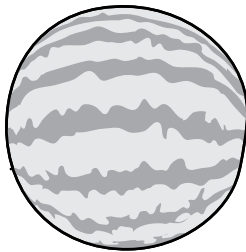


Saturn

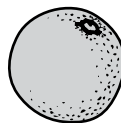


Titan

C



Sun



Saturn



Titan

D



Sun



Saturn



Titan

(continued on next page)

Standard: Develop and use a model to describe the role of gravity in the motions within galaxies and the solar system.

Cognitive Complexity:

SEP: High

DCI: Low

CCC: Medium

Overall Cognitive Complexity: Moderate Support

Rationale:

- A. All three objects are roughly the same size, which does not reflect the large differences in the sizes of the Sun, Saturn, and Titan.
- B. KEY: This comparison does the best job of showing the size difference between all the objects. The Sun is 12 times larger than Saturn and 277 times larger than Titan.**
- C. Although the watermelon is bigger than the orange, it is not enough larger to accurately reflect the difference in sizes between the Sun and Saturn. Also, the apple is almost the same size as the orange so a much smaller fruit should have been chosen.
- D. The three balls decrease in size; however, there is not much difference between the softball, baseball, and tennis ball. This does not do the best job of showing the big differences in the sizes of the Sun, Saturn, and Titan.

- 20** After viewing the models, the student claims that Titan only orbits Saturn and Saturn only orbits the Sun.

Which statement **best** evaluates the student's claim?

- A** The student is correct because only Saturn is shown orbiting the Sun in the model.
- B** The student is correct because all moons orbit planets and Titan is classified as a moon.
- C** The student is incorrect because all objects in the solar system orbit the Sun because it has the largest mass.
- D** The student is incorrect because Saturn has less mass than the Sun, which causes Titan to only orbit the Sun.

Standard: Develop and use a model to describe the role of gravity in the motions within galaxies and the solar system.

Cognitive Complexity:

SEP: Medium

DCI: Medium

CCC: Medium

Overall Cognitive Complexity: Moderate Support

Rationale:

- A. The student making the claim has not taken into account the limitations of the model or that the model is only showing one interaction.
- B. Although moons do orbit planets, they also orbit the Sun as the planet orbits the Sun.
- C. KEY: All objects in the solar system orbit the Sun, even if they orbit another object because the Sun's mass is so large that all objects within the solar system orbit its mass.**
- D. Saturn does have less mass than the Sun, but Saturn is closer to the Sun, which is why it orbits Saturn as well as the Sun.

This question has two parts. Be sure to answer both parts of the question.

21 Part A

Which sentence **best** compares the gravitational force between the pairs of objects?

- A** The gravitational force between Titan and Saturn is greater than the gravitational force between Titan and the Sun because of the masses of the interacting objects.
- B** The gravitational force between Titan and Saturn is greater than the gravitational force between Titan and the Sun because of the distances between the interacting objects.
- C** The gravitational force between Titan and Saturn is less than the gravitational force between Titan and the Sun because of the masses of the interacting objects.
- D** The gravitational force between Titan and Saturn is less than the gravitational force between Titan and the Sun because of the distances between the interacting objects.

Part B

Which sentence **best** compares the gravitational force between the pairs of objects?

- A** The gravitational force between the Sun and Saturn is greater than the gravitational force between the Sun and Titan because of the masses of the interacting objects.
- B** The gravitational force between the Sun and Saturn is greater than the gravitational force between the Sun and Titan because of the distances between the interacting objects.
- C** The gravitational force between the Sun and Saturn is less than the gravitational force between the Sun and Titan because of the masses of the interacting objects.
- D** The gravitational force between the Sun and Saturn is less than the gravitational forces between the Sun and Titan because of the distances between the interacting objects.

Standard: Develop and use a model to describe the role of gravity in the motions within galaxies and the solar system.

Cognitive Complexity:

SEP: Low

DCI: Medium

CCC: Medium

Overall Cognitive Complexity: Moderate Support

Rationale:

Part A

- A. The table shows that the combined masses of Titan and Saturn are smaller than the combined masses of Titan and the Sun.
- B. **KEY:** The orbital model shows that Titan orbits Saturn, which shows that the gravitational force between them is stronger. The decreased distance is why the gravitational force is stronger even though the table shows that the mass of the Sun is much larger than the mass of Saturn.
- C. The orbital model shows that Titan orbits Saturn, meaning that the gravitational attraction is greater than the gravitational attraction of the Sun and Titan. The table shows that the combined masses of Titan and Saturn are smaller than the combined masses of Titan and the Sun.
- D. The orbital model shows that Titan orbits Saturn, meaning that the gravitational attraction is greater than the gravitational attraction of the Sun and Titan.

Part B

- A. **KEY:** The data table shows that the average distance between Titan and Saturn to the Sun is the same. Because Saturn has more mass, there will be more gravitational attraction between Saturn and the Sun than Titan and the Sun.
- B. Distance between the interacting objects is not different according to the data table, so the reasoning behind the greater gravitational force is incorrect.
- C. Since Saturn and Titan are the same distance from the Sun, the masses of the interacting objects will have the effect on the difference in the gravitational forces. Gravitational force increases with mass, so Saturn and the Sun will have a greater gravitational force between them.
- D. Distance between the interacting objects is not different according to the data table, so the reasoning behind the greater gravitational force is incorrect. Since Saturn and Titan are the same distance from the Sun, the masses of the interacting objects will have the effect on the difference in the gravitational forces. Gravitational force increases with mass, so Saturn and the Sun will have a greater gravitational force between them.